MILITARY SPECIFICATION
ANTENNA, IFF, AS-2188( ) (3METERS) (10FEET)), AS-2189
(1.8 METERS (6 FEET))

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the performance of the IFF, AS-2188 (3 meters (10 feet)) and IFF, AS-2189 (1.8 meters (6 feet)) antennas with ISLS Interrogator Side Lobe Suppression (ISLS) capability, for installation aboard Naval vessels. An antenna consists of a directional IFF Antenna, whose halves may be fed separately and independently, a Hybrid Assembly and a Backfill Radiator.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited)

MILITARY

MIL-C-17/74	- Cable, Radio Frequency, Coaxial
MIL-S-901	- Shock Tests, H.I. (HIGH-IMPACT), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-16400	- Electronic Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification For
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods

PUBLICATIONS

MILITARY

MIL-HDBK-216	- RF Transmission Lines and Fittings

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5985

Beneficial comments (recommendations, additions, deletions) and any pertinent Data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with the requirements of MIL-E-16400 except as hereinafter specified. Screw threads and screw thread devices shall be in accordance with Handbook H28 (NBS).

3.2 First article sample. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.3).

3.3 Reliability

3.3.1 Quantitative reliability requirement. The equipment shall be designed to have a minimum acceptable mean-time-between-failure (MTBF) of 10,000 hours.

3.4 Design requirements.

3.4.1 Construction. The requirements of modular construction in MIL-E-16400 are not applicable. The antenna shall be constructed as one unit. The equipment shall be constructed to prevent internal accumulation of water or moisture that would adversely affect the RF performance.

3.4.2 Environmental service conditions.

3.4.2.1 Temperature. The equipment shall be designed to meet the Range 2 temperature requirements of MIL-E-16400, except that the upper limit shall be 150°C (302°F). The contractor shall test the operation of the antenna at this temperature.

3.4.2.2 Salt fog. The equipment shall be capable of withstanding the effects of the salt fog test specified herein. After completion of the test and cleaning, the base metal of the antenna shall not be visible through the finish, nor shall there be any evidence of blistering, softening, separation from the base metal, corrosion products, or other coating failure. The equipment shall maintain the specified performance after completion of the salt fog test.

3.4.2.3 Hydrostatic pressure. Hydrostatic pressure is not applicable.

3.4.3 Electrical design.

3.4.3.1 The radio frequency (RF) system of this equipment shall be capable of carrying pulses of RF energy of 10 kilowatts (kw) peak power at one percent duty cycle without breakdown, under service conditions.

3.4.3.2 Overload protection is not required.

3.4.3.3 Connectors and cables. RF type N connectors shall be used in this equipment and shall be selected from MIL-HDBK-216. Cables shall be in accordance with MIL-C-17/74.

3.4.3.4 RF feed system. The antennas shall have two RF input connectors to implement interrogation side lobe suppression. When fed from one input, a "sum" radiation pattern shall be generated. When fed from the other input, a "difference" pattern shall be generated. When using the "difference" input, a portion of the signal shall be routed from the "difference" input to the backfill radiator via a directional coupler, sufficient to exceed the "sum" pattern back lobe by at least 4.0 decibels (dB).

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3.4.4 Maintenance.

3.4.4.1 Test points and test features are not required.

3.4.5 Mechanical.

3.4.5.1 Shock requirements shall be for Grade A, Class I, lightweight, Type A of MIL-S-901.

3.4.5.1.1 The vibration requirements shall be Type I of MIL-STD-167.

3.4.5.1.1.1 The inclination requirements are not applicable.

3.4.5.2 Exposure. The equipment shall be designed to withstand the conditions incident to its exposed location on Naval vessels, and shall be watertight.

3.4.5.3 Materials and protection.

3.4.5.3.1 Hardware. All hardware shall be CRES 300 series, cadmium plated in accordance with QQ-P-416, Type II, Class 3.

3.4.5.3.2 Painting, undercoats. Except where application would degrade antenna performance, all exterior surfaces of the antenna and its coaxial conductors shall be spray-painted prior to assembly with the following, or its equivalent:

	Coating	Dry Film Thickness
1st coat:	Devran 201 Lt. Green	2-3 mils
2nd coat:	Devran 201 Buff	2-3 mils
3rd coat:	Devran 209 Haze Gray No. 27	2-3 mils

Three coats shall be applied to each part unless otherwise specified.

3.4.5.3.3 Painting, finish. After assembly and testing, the complete antenna assembly shall be spray-painted with one coat (2-3 mil thickness) of Devran 209 Haze Gray No. 27, except where application will degrade antenna performance.

3.5 Electrical requirements

3.5.1 AS-2188 ( ) antenna. The AS-2188 ( ) antenna shall have the following electrical characteristics over the frequency range of 1030 ±10 megahertz (MHz) when energized through the feed system and pedestal:

(a) Vertical polarization.
(b) A mean impedance of 51 ohms nominal (both inputs).
(c) A voltage standing wave ratio not greater than 3.6 dB at the input terminals to the antenna.
(d) Gain of at lease 18.5 dB referenced to an isotropic radiator.
(e) A horizontal sum pattern with main lobe beam width of not more than 7.3 degrees at 3 dB below maximum and not more than 20 degrees at 20 dB below maximum of the main lobe. Maximum of the side lobes shall be at least 26 dB below the maximum of the main lobe, and the maximum of the back lobe shall be at least 26 dB below the maximum of the main lobe.
(f) The horizontal "difference" pattern shall cover all side lobes and back lobes by at least 4.0 dB. The depth of the hull at 0° (the "sum" pattern main beam) shall be at least 30 dB below the "sum" pattern main beam maximum point. The "difference" pattern shall be designed to obtain the greatest degree of "beam" sharpening" possible.

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(g) The antenna shall have a vertical sum pattern as follows:
(1) Total beam width, as measured at the points 3 dB down from the peak of the main beam, of at least 50°.
(2) At ±40° and -20° elevation, no more than 5 dB down from the peak of the main beam.
(3) At ±60° elevation, no more than 12 dB down from the peak of the main beam.
(4) At ±90° elevation, at least 18 dB down from the peak of the main beam.
(5) At ±145° elevation, at least 30 dB down from the peak of the main beam.
(6) At -80° elevation, at least 24 dB down from the peak of the main beam.
(7) From ±150° through ±180° -130°, elevation, the back lobe shall be at least 30 dB below the peak of the main beam.

3.5.1.1 The antennas shall be designed for optimum performance at 1030 MHz and at 1090 MHz.

3.5.2 AS-2189 antenna. The AS-2189 antenna shall have the following electrical characteristics over the frequency range of 1030 ±10 and 1090 ±10 MHz when energized through the feed system and pedestal:

(a) Vertical polarization.
(b) A mean impedance of 51 ohms (both inputs).
(c) A power standing wave ratio not greater than 3.5 dB at the input terminal to the antenna.
(d) Gain of at least 16.5 dB referenced to an isotropic radiator.
(e) A horizontal "sum" pattern with main lobe beam width of no more than 11 degrees at 3 dB below maximum, and no more than 25 degrees at 20 dB below maximum of main lobe. Maximum of the side lobes shall be at least 26 dB below the maximum of the main lobe, and the maximum back lobe shall be at least 30 dB below the main lobe.
(f) The horizontal "difference pattern" shall cover all side lobes and back lobes by at least 4.0 dB. The depth of the null at 0° (the "sum" pattern main beam) shall be at least 30 dB below the "sum" pattern main beam maximum point. The "difference" pattern shall be designed to obtain the greatest degree of "beam sharpening" possible.
(g) The antenna shall have a vertical sum pattern as follows:
(1) Total beam width, as measured at the points 3 dB down from the peak of the main beam, of at least 50°.
(2) At +40° and -20° elevation, no more than 5 dB down from the peak of the main beam.
(3) At +60° elevation, no more than 12 dB down from the peak of the main beam.
(4) At ±90° elevation, at least 18 dB down from the peak of the main beam.
(5) At ±145° elevation, at least 30 dB down from the peak of the main beam.
(6) At -80° elevation, at least 24 dB down from the peak of the main beam.
(7) From ±150° through ±180° to -130°, elevation, back lobe shall be at least 30 dB below the peak of the main beam.

3.6 Design requirements.

3.6.1 The design of the antenna shall be such that when it is not mounted it shall be possible to feed the antenna directly by coaxial cables through RF connectors, as described in 3.4.3.4. The electrical characteristics of the antenna and feed system shall not be adversely affected when the antenna is fed in this manner.

3.6.1.1 The RF coupling and mechanical mounting provisions for mounting the antenna on the pedestal shall be such that the antenna can, without modification, be mounted on the AN/UPA-57 pedestal and on the pedestal for the AN/UPA-22 antenna system, as shown in figure 1.

3.6.2 Both antennas shall incorporate a hybrid duplexer with a directional coupler for the backfill radiator, and a backfill radiator to implement ISLS operation. The hybrid duplexer and directional coupler shall comprise the Hybrid Assembly, which shall be made integral with the antenna, utilizing stripline techniques.

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3.6.2.1 Hybrid duplexer. The hybrid duplexer shall be capable of operation with the RF pulses specified in 3.4.3.1 and having the following characteristics:

(a) Power - 10 kw at 1.0 percent duty cycle.
(b) Frequency - 1030 ±10 and 1090 ±0.1 microsecond (µsec).
(c) Width - 0.8 ±0.1 µsec and 0.50 ±0.1 (µsec).
(d) Rise time - 0.10 µsec or less.
(e) Decay time - 0.15 µsec or less.
(f) Impedance - 51 ±2 ohms.
(g) Minimum separation, leading edge 1.0 ±0.1 µsec.
(h) Power balance - 0.4 dB or less.
(i) Phase tolerance - Sum 0 ±4 degrees.

3.6.2.2 Directional coupler. A coupler incorporating a 6 ±1 dB attenuation shall be provided as an integral part of the hybrid assembly. Feed to the backfill antenna shall be via a Type N/HN connector. The six dB coupler shall have a characteristic impedance of 51 ±2 ohms.

3.6.3 The antenna base and mount assembly shall be an integral part of the antenna. The base shall be basically a one-piece construction, symmetrical about the antenna center-line, and extending to the center of each moment arm of the antenna. Two sets of mounting holes, spaced as shown in figure 2 shall be provided. The design of the antenna shall be such that it will be possible to mount the antenna on a radar antenna structure when proper mounting fixtures are provided: Mounting provision shall be made on the antenna for an RF Switch (not part of this specification), no larger than 26 cm X 15.5 cm X 3.9 cm (10 inches X 6 inches X 1 1/2 inches) and weighing no more than 2.3 kg (5 pounds). The location chosen shall require the approval of the design agency.

3.6.4 The size of the AS-2188 ( ) antenna shall not exceed 300 cm (112 inches) in length by 48.5 cm (19 inches) in height, and the size of the AS-2189( ) antenna shall not exceed 190 cm (74 inches) in length and 48.5 cm (19 inches) in height. The depth of the array including bracing structure shall be a minimum consistent with good design practices and subject to the approval of the command or agency concerned.

3.6.5 The weight of the AS-2188( ) antenna assembly shall not exceed 33.5 kg (74 pounds) and the weight of the AS-2189( ) antenna assembly 25 kg (64 pounds).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered product to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality Assurance Terms and Definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection
(b) Quality conformance inspection
    (1) Production inspection
    (2) Production control inspection
    (3) Environmental inspection
(c) Inspection of preparation for delivery

4.3 First article inspection. Unless otherwise specified (see 6.2.1) one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I and 4.3.1.

TABLE I. First Article Inspection

Examination or Test	Requirement paragraph	Test Method paragraph
Surface examination		4.5.1
Performance test	3.6.1	4.5.2
Sunshine	3.4.5.2	4.5.3
Enclosure	3.4.1, 3.4.5.2	4.5.4
Fungus		4.5.12
Temperature	3.4.2.1	4.5.5
Humidity		4.5.6
Power		4.5.7
Shock and vibration	3.4.5.1, 3.4.5.1.1	4.5.8
Salt fog	3.4.2.2	4.5.9
Combined wind speed & icing test		4.5.10
Interchangeability	3.6.1.1, 3.6.3	4.5.11

4.3.1 Verification of characteristics. In addition, after first article tests have been performed, the following characteristics of the antennas (previously determined) shall be checked to determine compliance with the requirements of this specification:

Impedance (see 3.5.1(b) and 3.5.2(b))
VSWR (see 3.5.1(c) and 3.5.2(c))
Gain (see 3.5.1(d) and 3.5.2(d))
Pattern (see 3.5.1(e), (f), (g) and 3.5.2(e), (f), (g))
Power Capacity (see 3.4.3.1))
RF Feed System (see 3.4.3.4)

4.4 Quality conformance inspection

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the requirements of table II.

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Table II. Production Inspection

Surface Examination	Requirement Paragraph	Test Method Paragraph
Surface Examination		4.5.1
Performance Test	3.6.1	4.5.2

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III below and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection utilizing an AQL of 6.5 percent. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table III. Production Control Inspection

Examination or Test	Requirement Paragraph	Test Method Paragraph
Sunshine	3.4.5.2	4.5.3
Enclosure	3.4.1, 3.4.5.2	4.5.4
Power	3.4.3.1	4.5.7
Interchangeability	3.6.1.1, 3.6.3	4.5.11

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to, and have passed, production control inspections may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all visible damage.

4.4.3 Environmental tests. Environmental tests shall be as listed in table IV and shall be performed on units which have been subjected to and passed production inspection.

Table IV. Environmental Tests

Examination or Test	Requirement Paragraph	Test Method Paragraph
Temperature	3.4.2.1	4.5.5
Humidity		4.5.6
Shock and Vibration	3.4.5.1, 3.4.5.1.1	4.5.8
Salt Fog	3.4.2.2	4.5.9

4.4.3.1 In addition, the verification of the characteristics of the antenna as specified in 4.3.1 shall be made after environmental testing.

4.4.3.2 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments produced or fraction thereof. The first sample equipment shall be selected from the first month's production.

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4.4.3.3 Nonconforming environmental sample units. If a sample unit fails the inspection specified in paragraph 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes. If the Quality Assurance Representative does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3.4 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to, and have passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Inspection Procedures

4.5.1 Surface examination. Each antenna assembly shall be examined for workmanship, assembly and fit, mechanical mounting, electrical connections, materials, parts, finish and treatment for prevention of corrosion to determine that workmanship requirements of MIL-STD-454 and material and parts requirements of this specification have been met. Particular attention shall be given to the following:

(a) Completeness
(b) Cable runs including RF connectors, couplers, and heat-shrinkable boots covering RF connectors
(c) Painting and finish
(d) Antenna mounting provisions
(e) Weight and size
(f) Marking
(g) Other visual defects

4.5.2 Performance test. The performance test shall be in accordance with MIL-E-16400 to verify compliance with 3.6.1. As a minimum the following antenna characteristics shall be measured over the frequency range of 1030 ±10 and 1090 ±10 MHz to verify compliance with electrical requirements of this specification.

Characteristic	Requirement Paragraph
Impedance	3.5.1(b), 3.5.2(b)
VSWR	3.5.1(c), 3.5.2(c)
Gain	3.5.1(d), 3.5.2(d)
Pattern	3.5.1(e), (f), (g), 3.5.2(e), (f), (g)
Power Capacity	3.4.3.1
RF Feed System	3.4.3.4 and 3.5.1

4.5.3 Sunshine. The equipment shall be subjected to the sunshine tests specified in Method 505, Procedure II of MIL-STD-810 to verify compliance with 3.4.5.2.

4.5.4 Enclosure test. The equipment shall be subjected to the watertight test of MIL-STD-108 to verify compliance with 3.4.1 and 3.4.5.2.

4.5.5 Temperature test. The antenna assembly shall be subjected to temperature operating tests (Range 2) in accordance with MIL-E-16400, except that the upper temperature limit shall be 150°C, to verify compliance with 3.4.2.1.

4.5.6 Humidity test. Humidity testing shall be performed in accordance with MIL-E-16400. Prior to and following temperature and humidity cycling tests, performance parameters listed in 4.3.1 shall be measured and verified.

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4.5.7 Power. Each antenna assembly shall be tested to verify the ability of the antenna to transmit RF energy of at least 10 kw peak power at one percent duty cycle, without breakdown, under service conditions.

4.5.8 Shock and vibration. The antennas shall be subjected to shock and vibration testing in accordance with MIL-E-16400 to demonstrate compliance with 3.4.5.1 and 3.4.5.1.1. A simulated load shall be used for the RF switch. At the conclusion of these tests, the characteristic listed in 4.3.1 shall be verified and the antenna visually examined for damage or distortion.

4.5.9 Salt fog. The completed equipment shall be subjected to the salt fog test specified in Method 509 of MIL-STD-810.

4.5.10 Wind speed and icing test. In addition to the weight of the antenna, the antenna shall withstand the following simulated load conditions, simultaneously applied:

(a) Winds having a relative velocity of 100 knots (non-operating)
(b) Ice loads of 2.1 kg (4.5 pounds) per square foot

4.5.11 Inspection of interchangeability. The dimensions specified in figures 1 and 2 shall be gaged or measured to determine conformance to the requirements of 3.6.1.1 and 3.6.3:

(a) Alignment hole diameter and mounting hole diameter (see figures 1 and 2)
(b) Antenna pedestal mounting box hole pattern and antenna base mounting hole pattern (see figures 1 and 2)

4.5.12 Fungus. The antenna shall be subjected to the fungus test specified in Method 508 of MIL-STD-810.

4.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for use aboard naval vessels.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements

(a) Title, number, and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Marking, levels of preservation and packaging and packing required (see 5.1).

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6.3 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Date Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification include but are not limited to the following:

Paragraph	Data Requirement	Applicable DID
4.3	First Article Inspection Procedure	DI-T-4901
4.4.3	Environmental Test Plan/Procedures	DI-T-37081 T-108-2

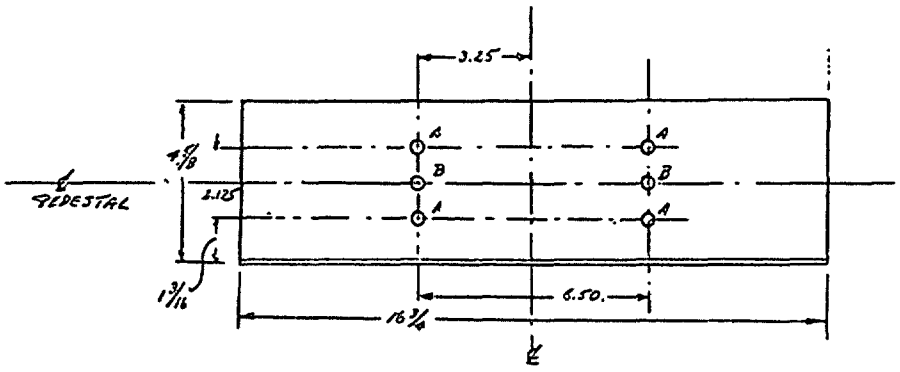
(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.4 First article. Invitations for bids should provide that the government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending procurement.

6.5 Initial shipment. Initial shipment of end items will be contingent upon successful completion of the environmental inspection.

Preparing activity:
Navy-EC
(Project 5985-N475)

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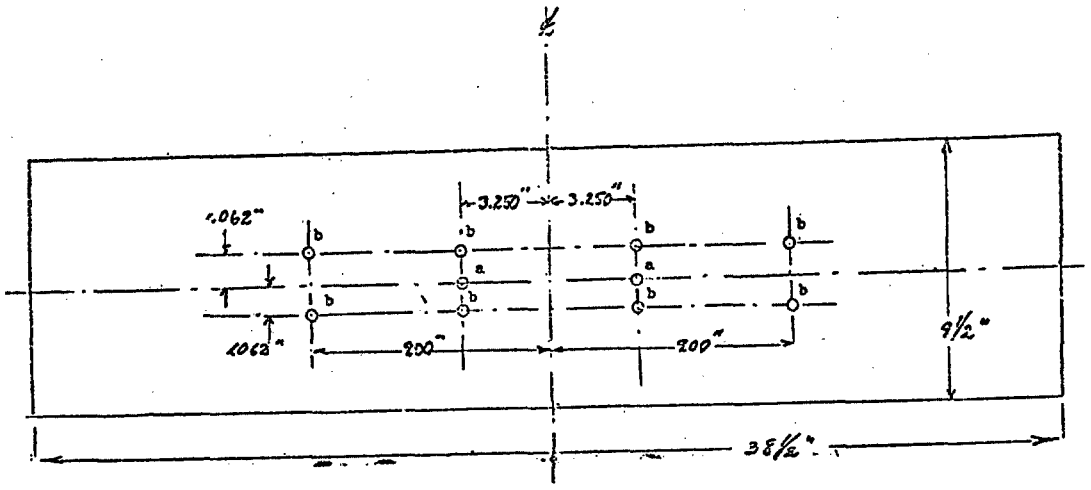


A - 3/8-24 CAPTIVE SELF-LOCK NUTS (4 HOLES)
B - 10 MMD ALIGNMENT HOLES (2 HOLES) HOLE TOLERANCES - ±0.4 MM

FIGURE 1. AT-352/353 antenna mounting box

Inches	Centimeters
4 5/8	11.7475
1 3/16	3.01625
2.125	5.3975
3.25	8.255
16 3/4	42.545
6.5	16.51*

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NOTES:
(a) GUIDE PIN HOLES - 1 MMD (2)
(b) 3/8" x 24 CAPTIVE SELF LOCKING NUTS (8)
(c) HOLE TOLERANCE - 0.4 mm

FIGURE 2. Mounting bracket - directional IFF antenna (typical)

Inches	Millimeters
1.062	26.9748
9	228.6
3.250	82.55
9 1/2	241.3
38 1/2	977.9

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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